Numerical Analysis

Chapter 4: Approximation theory

Department of Mathematics & Statistics
Missouri University of Science & Technology

Textbook: Classical and Modern Numerical Analysis:
Theory, Methods and Practice
A. S. Ackleh, E. J. Allen, R. B. Kearfott and P. Seshaiyer
CRC Press, Boca Raton, FL, 2010

Outline

1. Introduction
2. Best approximation
3. Polynomial Approximation
4. Piecewise polynomial approximation
5. Other topics

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3 Polynomial Approximation

4 Piecewise polynomial approximation

5 Other topics

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Definition (Span)

Let $u_1, u_2, \cdots, u_n \in V$. The set of all linear combination of $u_1, u_2, \cdots, u_n$ is called the span of $u_1, u_2, \cdots, u_n$, which is denoted by $span\{u_1, u_2, \cdots, u_n\}$

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- P^1 is a subspace of P^2 and P^j $(j = 0, 1, \cdots, k)$ is a subspace of P^k.
- $P^k[a, b]$ is a subspace of $C[a, b]$.
- If $W = span\{u_1, u_2, \cdots, u_n\}$ and $u_1, u_2, \cdots, u_n \in V$, then it is easy to verify that W is a subspace of V.

Vector space

Definition (Linearly independent/dependent)

Let V be a vector space. Then $u_1, u_2, \cdots, u_n \in V$ are linearly independent if $\sum_{i=1}^{n} \alpha_i u_i = 0$ implies that $\alpha_i = 0$ $(i = 1, 2, \cdots, n)$.
Otherwise, $u_1, u_2, \cdots, u_n \in V$ are linearly dependent.

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Example

Consider $V = C[0, 1]$. $u_1 = 1$, $u_2 = x$ and $u_3 = x^2$ are linearly independent, but $u_1 = 1$, $u_2 = x$ and $u_3 = 2 - 3x$ are linearly dependent.

Vector space

Definition (Basis)

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Definition (Dimension)

If V has a basis of n elements, Then, n is called the dimension of V.

Vector space

Lemma (I)

If V is a n-dimensional vector space and $u_1, u_2, \cdots, u_n \in V$ are linearly independent, then $V = span\{u_1, u_2, \cdots, u_n\}$. Moreover, every basis of V has n elements and any collection of $n+1$ elements in V is linearly dependent.

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If V is a n-dimensional vector space and $u_1, u_2, \cdots, u_n \in V$ are linearly independent, then $V = span\{u_1, u_2, \cdots, u_n\}$. Moreover, every basis of V has n elements and any collection of $n+1$ elements in V is linearly dependent.

Example

$P^2 = span\{1, x, x^2\}$. Since 1, x, and x^2 are linearly independent, they form a basis of P^2 and the dimension of P^2 is 3.

Norm

Definition (Norm of a function)

Assume V is a vector space. For any function $v \in V$, a norm of v is defined to be a number $\|v\|$ satisfying

1. $\|v\| \geq 0$.
2. $\|v\| = 0$ if and only if $v = 0$.
3. $\|\lambda v\| = |\lambda| \, \|v\|$ for $\lambda \in \mathbb{R}$ or$\mathbb{C}$.
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- $\|v\|_1 = \int_a^b |v(x)|\,\rho(x)\,dx$ is called the L^1 norm with weight function $\rho(x) > 0$. When $\rho(x) = 1$, it is called the L^1 norm.

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- *A geometric explanation of the best approximation: see Figure 4.1 on page 193 of the textbook.*

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- *Yes!*

Existence of a best approximation

Theorem (I)

Let W be an $n+1$-dimensional subspace of a normed linear space V. Let $u_0, u_1, \cdots, u_n$ be linearly independent elements of W. That is, $W = span\{u_0, u_1, \cdots, u_n\}$. Then there is a $p \in W$, i.e., $p = \sum_{j=0}^{n} \alpha_j u_j$ for a given $f \in V$, such that

$$\|f - p\| = \left\| f - \sum_{j=0}^{n} \alpha_j u_j \right\| = \min_{\gamma_0, \gamma_1, \cdots, \gamma_n} \left\| f - \sum_{j=0}^{n} \gamma_j u_j \right\|$$

That is, $\|f - p\| \leq \|f - q\|$ for all $q \in W$. Hence p is the best approximation in W to $f \in V$ with respect to norm $\|\cdot\|$.

Existence of a best approximation

Proof.

See pages 192-194 of the textbook. (Independent study problem) □

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  Since $f(x) = e^x - p$ is a monotonic function of x, $\max\limits_{0\le x\le 1} |e^x - p|$ is the larger one of $\left|e^0 - p\right| = |1 - p|$ and $|e - p|$.

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  The middle point $p = \frac{1}{2}(e + 1)$.

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- So the minimum value of the larger one of the two distances is $\frac{1}{2}(e-1)$. So is the minimum value of $\max\limits_{0 \le x \le 1} |e^x - p|$.

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  Then

$$\begin{aligned} g'(p) &= \int_0^1 \frac{d\,(e^x - p)^2}{dp}\ dx \\ &= \int_0^1 -2\,(e^x - p)\ dx \\ &= -2(e - p - 1). \end{aligned}$$

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Remark

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- *Yes! The best approximation in inner product spaces.*

Preparation: Inner product space

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- Can you see any relationship between this inner product and the L^2 norm?
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Theorem (II)

Any real inner product space V is a real normed linear space with norm defined by $\|v\| = (v, v)^{\frac{1}{2}}$.

Preparation: more topics

- Complex inner product spaces.
- Hilbert and Banach spaces.
- Cauchy-Schwarz inequality.
- .....

Definition of the best approximation in inner product spaces

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Let $W = span\{w_1, w_2, \cdots, w_n\}$ be a finite-dimensional subspace of an inner product space V. Here $\{w_i\}_{i=1}^n$ is a linearly independent set. Given $v \in V$, a best approximation in W to v with respect to the norm $\|v\| = (v, v)^{\frac{1}{2}}$ is a $w \in W$ such that the distance $\|v - w\|$ is the least among all $w \in W$. That is, $\|v - w\| \leq \|v - u\|$ for any $u \in W$.

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- *But how?*

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- Since $W = span\{w_1, w_2, \cdots, w_n\}$, then $w = \sum_{j=1}^{n} \alpha_j w_j$ for some α_j $(j = 1, 2, \cdots, n)$.

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- Question: how to find these α_j $(j = 1, 2, \cdots, n)$?

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&= \left(\sum_{j=1}^{n} \alpha_j w_j - v, \sum_{k=1}^{n} \alpha_k w_k - v \right) \\
&= \sum_{j=1}^{n} \sum_{k=1}^{n} \alpha_j \alpha_k (w_j, w_k) - \sum_{j=1}^{n} \alpha_j (v, w_j) \\
&\quad - \sum_{k=1}^{n} \alpha_k (v, w_k) + (v, v)
\end{aligned}
$$

Derivation of the best approximation in inner product spaces

- By $\|v\| = (v, v)^{\frac{1}{2}}$, we have

$$
\begin{aligned}
\|w - v\|^2 &= (w - v, w - v) \\
&= \left(\sum_{j=1}^{n} \alpha_j w_j - v, \sum_{k=1}^{n} \alpha_k w_k - v \right) \\
&= \sum_{j=1}^{n} \sum_{k=1}^{n} \alpha_j \alpha_k (w_j, w_k) - \sum_{j=1}^{n} \alpha_j (v, w_j) \\
&\quad - \sum_{k=1}^{n} \alpha_k (v, w_k) + (v, v) \\
&= F(\alpha_1, \alpha_2, \cdots, \alpha_n).
\end{aligned}
$$

Derivation of the best approximation in inner product spaces

- Thus, the problem reduces to finding the minimum of F as a function of $\alpha_1, \alpha_2, \cdots, \alpha_n$.

Derivation of the best approximation in inner product spaces

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$$\frac{\partial \left(\sum_{j=1}^{n} \alpha_j (v, w_j) \right)}{\partial \alpha_i} = (v, w_i)$$

$$\frac{\partial \left(\sum_{k=1}^{n} \alpha_k (v, w_k) \right)}{\partial \alpha_i} = (v, w_i)$$

$$\frac{\partial (v, v)}{\partial \alpha_i} = 0.$$

Derivation of the best approximation in inner product spaces

- As for $\sum_{j=1}^{n}\sum_{k=1}^{n}\alpha_j\alpha_k(w_j,w_k)$, we have

$$\begin{aligned}&\sum_{j=1}^{n}\sum_{k=1}^{n}\alpha_j\alpha_k(w_j,w_k)\\ =&\sum_{j=1,j\neq i}^{n}\sum_{k=1,k\neq i}^{n}\alpha_j\alpha_k(w_j,w_k)+\sum_{k=1,k\neq i}^{n}\alpha_i\alpha_k(w_i,w_k)\\ &+\sum_{j=1,j\neq i}^{n}\alpha_j\alpha_i(w_j,w_i)+\alpha_i^2(w_i,w_i).\end{aligned}$$

Derivation of the best approximation in inner product spaces

- Hence

$$\frac{\partial\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\alpha_j\alpha_k(w_j,w_k)\right)}{\partial\alpha_i}$$

$$= 0+\sum_{k=1,k\neq i}^{n}\alpha_k(w_i,w_k)+\sum_{j=1,j\neq i}^{n}\alpha_j(w_j,w_i)+2\alpha_i(w_i,w_i)$$

$$= 2\sum_{j=1}^{n}\alpha_j(w_j,w_i).$$

Derivation of the best approximation in inner product spaces

- Then

$$\frac{\partial F}{\partial \alpha_i} = 2\sum_{j=1}^{n} \alpha_j (w_j, w_i) - 2(v, w_i).$$

Derivation of the best approximation in inner product spaces

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$$\frac{\partial F}{\partial \alpha_i} = 2\sum_{j=1}^{n} \alpha_j (w_j, w_i) - 2(v, w_i).$$

- Setting $\frac{\partial F}{\partial \alpha_i} = 0$, then

$$\sum_{j=1}^{n} \alpha_j (w_j, w_i) = (v, w_i), \text{ for } i = 1, 2, \cdots, n.$$

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- Then

$$\frac{\partial F}{\partial \alpha_i} = 2\sum_{j=1}^{n} \alpha_j (w_j, w_i) - 2(v, w_i).$$

- Setting $\frac{\partial F}{\partial \alpha_i} = 0$, then

$$\sum_{j=1}^{n} \alpha_j (w_j, w_i) = (v, w_i), \text{ for } i = 1, 2, \cdots, n.$$

- Given v and w_i $(i = 1, 2, \cdots, n)$, we can solve the above linear system to obtain α_j $(j = 1, 2, \cdots, n)$, hence $w = \sum_{j=1}^{n} \alpha_j w_j$.

Another way to understand the best approximation in inner product spaces

Definition (Projection)

Let W be a finite-dimensional subspace of an inner product space V. An operator P that maps V into W such that $P^2 = P$ is called a projection operator from V into W.

Another way to understand the best approximation in inner product spaces

Remark

Assume $W = span\{w_1, w_2, \cdots, w_n\}$ *where* $\{w_i\}_{i=1}^n$ *is a linearly independent set. Then define* $P : V \to W$ *as* $Pv = \sum_{j=1}^{n} \alpha_j w_j$ *where the coefficients* α_j $(j = 1, 2, \cdots, n)$ *satisfy*

$$\sum_{j=1}^{n} \alpha_j (w_j, w_i) \;=\; (v, w_i), \;\; \text{for } i = 1, 2, \cdots, n.$$

Then the projection Pv *of* v *is actually the best approximation in* W *to* v *with respect to the norm* $\|v\| = (v, v)^{\frac{1}{2}}$.

Further topics of the best approximation in inner product spaces

Remark

- *How to solve the above linear system?*

Further topics of the best approximation in inner product spaces

Remark

- *How to solve the above linear system?*
- *Chapter 3!*

Further topics of the best approximation in inner product spaces

Remark

- *How to solve the above linear system?*
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- *Is it possible to find some special basis* w_i $(i = 1, 2, \cdots, n)$ *to dramatically simplify the above linear system?*

Further topics of the best approximation in inner product spaces

Remark

- *How to solve the above linear system?*
- *Chapter 3!*
- *Is it possible to find some special basis* w_i $(i = 1, 2, \cdots, n)$ *to dramatically simplify the above linear system?*
- *Yes! Orthogonal basis* w_i $(i = 1, 2, \cdots, n)$*!*

Orthogonal basis

Definition (Orthogonal/orthonormal)

Let V be an inner product space. Two vectors u and v in V are called orthogonal if $(u, v) = 0$. A set of such vectors that are pairwise orthogonal is called orthonormal if $(u, u) = 1$ for each u in that set.

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Definition (Orthogonal/orthonormal basis)

Let $W = span\{w_1, w_2, \cdots, w_n\}$ be a finite-dimensional subspace of an inner product space V. If $w_1, w_2, \cdots, w_n$ are orthogonal (or orthonormal), then they form an orthogonal (or orthonormal) basis of W.

Orthogonal basis

Remark

- *If* $w_1, w_2, \cdots, w_n$ *are pairwise orthogonal, then*

$$(w_j, w_i) = \delta_{ji} = \begin{cases} 0, & \text{if } j \neq i, \\ 1, & \text{if } j = i. \end{cases}$$

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Hence

$$\sum_{j=1}^{n} \alpha_j (w_j, w_i) = (v, w_i), \ \ i = 1, 2, \cdots, n,$$

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Hence

$$\sum_{j=1}^{n} \alpha_j (w_j, w_i) = (v, w_i), \ \ i = 1, 2, \cdots, n,$$

$$\Rightarrow \quad \alpha_i = \frac{(v, w_i)}{(w_i, w_i)}, \ \ i = 1, 2, \cdots, n,$$

Orthogonal basis

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$$\Rightarrow \quad \alpha_i = \frac{(v, w_i)}{(w_i, w_i)}, \ \ i = 1, 2, \cdots, n,$$

$$\Rightarrow \quad w = Pv = \sum_{j=1}^{n} \frac{(v, w_j)}{(w_j, w_j)} w_j.$$

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Orthogonal basis

Theorem (III)

Let $W = span\{w_1, w_2, \cdots, w_n\}$ be a finite-dimensional subspace of an inner product space V. If $w_1, w_2, \cdots, w_n$ are orthonormal, then $w = Pv = \sum_{j=1}^{n} (v, w_j) w_j$ is the best approximation in W to v with respect to the norm $\|v\| = (v, v)^{\frac{1}{2}}$. That is, $\|v - Pv\| \leq \|v - u\|$ for any $u \in W$.

Orthogonal basis

Theorem (III)

Let $W = span\{w_1, w_2, \cdots, w_n\}$ be a finite-dimensional subspace of an inner product space V. If $w_1, w_2, \cdots, w_n$ are orthonormal, then $w = Pv = \sum_{j=1}^{n} (v, w_j) w_j$ is the best approximation in W to v with respect to the norm $\|v\| = (v, v)^{\frac{1}{2}}$. That is, $\|v - Pv\| \leq \|v - u\|$ for any $u \in W$.

Proof.

See pages 199-201 of the textbook for the proof and the related materials, such as orthogonal complement and the orthogonal decomposition. (Independent study problem) □

Orthogonal basis

Remark

- *Usually we can easily obtain a basis* $\{u_1, u_2, \cdots, u_n\}$ *which are linearly independent.*

Orthogonal basis

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- *Question: how do we find an orthonormal (or at least orthogonal) basis* $\{w_1, w_2, \cdots, w_n\}$*?*

Orthogonal basis

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- *Usually we can easily obtain a basis* $\{u_1, u_2, \cdots, u_n\}$ *which are linearly independent.*
- *Question: how do we find an orthonormal (or at least orthogonal) basis* $\{w_1, w_2, \cdots, w_n\}$*?*
- *Answer: Gram-Schmidt process!*

Orthogonal basis

Gram-Schmidt process:

1. $v_1 = u_1$

Orthogonal basis

Gram-Schmidt process:

1. $v_1 = u_1$
2. $v_j = u_j - \sum_{k=1}^{j-1} \frac{(u_j, v_k)}{(v_k, v_k)} v_k$ for $j = 2, 3, \cdots, m$.

Orthogonal basis

Gram-Schmidt process:

1. $v_1 = u_1$
2. $v_j = u_j - \sum_{k=1}^{j-1} \frac{(u_j, v_k)}{(v_k, v_k)} v_k$ for $j = 2, 3, \cdots, m$.
3. $w_j = \frac{v_j}{\|v_j\|}$ for $j = 1, 2, 3, \cdots, m$.

Orthogonal basis

Theorem (IV)

$\{v_1, v_2, \cdots, v_n\}$ is an orthogonal basis and $\{w_1, w_2, \cdots, w_n\}$ is an orthonormal basis. Moreover, $span\{u_1, u_2, \cdots, u_n\} = span\{v_1, v_2, \cdots, v_n\} = span\{w_1, w_2, \cdots, w_n\}$.

Orthogonal basis

Theorem (IV)

$\{v_1, v_2, \cdots, v_n\}$ is an orthogonal basis and $\{w_1, w_2, \cdots, w_n\}$ is an orthonormal basis. Moreover, $span\{u_1, u_2, \cdots, u_n\} = span\{v_1, v_2, \cdots, v_n\} = span\{w_1, w_2, \cdots, w_n\}$.

Proof.

See page 202 of the textbook. (Independent study problem) □

Orthogonal basis

- Example 3: Let $V = C[-1, 1]$, $W = span\{1, x, x^2\}$, $(f, g) = \int_{-1}^{1} f(x)g(x)\, dx$ for $f, g \in V$, and $\|f\| = (f, f)^{\frac{1}{2}}$. What are the orthonormal basis $\{w_1, w_2, w_3\}$ of W? Find the best approximation in W to $f(x) = e^x \in V$.

Orthogonal basis

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- Solution:

$$v_1 = u_1 = 1,$$

Orthogonal basis

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- Solution:

$$\begin{aligned} v_1 &= u_1 = 1, \\ \|v_1\|^2 &= (v_1, v_1) = \int_{-1}^{1} 1\,dx = 2, \end{aligned}$$

Orthogonal basis

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- Solution:

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Orthogonal basis

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- Solution:

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\begin{aligned}
v_1 &= u_1 = 1, \\
\|v_1\|^2 &= (v_1, v_1) = \int_{-1}^{1} 1\,dx = 2, \\
w_1 &= \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}}, \\
v_2 &= u_2 - \frac{(u_2, v_1)}{(v_1, v_1)} v_1 = x - \frac{1}{2}\int_{-1}^{1} x\,dx = x,
\end{aligned}
$$

Orthogonal basis

- Example 3: Let $V = C[-1,1]$, $W = span\{1, x, x^2\}$, $(f,g) = \int_{-1}^{1} f(x)g(x)\,dx$ for $f, g \in V$, and $\|f\| = (f,f)^{\frac{1}{2}}$. What are the orthonormal basis $\{w_1, w_2, w_3\}$ of W? Find the best approximation in W to $f(x) = e^x \in V$.
- Solution:

$$
\begin{aligned}
v_1 &= u_1 = 1, \\
\|v_1\|^2 &= (v_1, v_1) = \int_{-1}^{1} 1\,dx = 2, \\
w_1 &= \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}}, \\
v_2 &= u_2 - \frac{(u_2, v_1)}{(v_1, v_1)} v_1 = x - \frac{1}{2}\int_{-1}^{1} x\,dx = x, \\
\|v_2\|^2 &= (v_2, v_2) = \int_{-1}^{1} x^2\,dx = 2/3,
\end{aligned}
$$

Orthogonal basis

- Continued solution:

$$w_2 = \frac{v_2}{\|v_2\|} = \frac{x}{\sqrt{2/3}},$$

Orthogonal basis

- Continued solution:

$$
\begin{aligned}
w_2 &= \frac{v_2}{\|v_2\|} = \frac{x}{\sqrt{2/3}}, \\
v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2
\end{aligned}
$$

Orthogonal basis

- Continued solution:

$$
\begin{aligned}
w_2 &= \frac{v_2}{\|v_2\|} = \frac{x}{\sqrt{2/3}}, \\
v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 \\
&= x^2 - \frac{1}{2} \int_{-1}^{1} x^2 \, dx - \frac{x}{2/3} \int_{-1}^{1} x^3 \, dx
\end{aligned}
$$

Orthogonal basis

- Continued solution:

$$
\begin{aligned}
w_2 &= \frac{v_2}{\|v_2\|} = \frac{x}{\sqrt{2/3}}, \\
v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 \\
&= x^2 - \frac{1}{2} \int_{-1}^{1} x^2 \, dx - \frac{x}{2/3} \int_{-1}^{1} x^3 \, dx \\
&= x^2 - \frac{1}{2} \cdot \frac{2}{3} - \frac{x}{2/3} \cdot 0 = x^2 - \frac{1}{3}
\end{aligned}
$$

Orthogonal basis

- Continued solution:

$$
\begin{aligned}
w_2 &= \frac{v_2}{\|v_2\|} = \frac{x}{\sqrt{2/3}}, \\
v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 \\
&= x^2 - \frac{1}{2} \int_{-1}^{1} x^2 \, dx - \frac{x}{2/3} \int_{-1}^{1} x^3 \, dx \\
&= x^2 - \frac{1}{2} \cdot \frac{2}{3} - \frac{x}{2/3} \cdot 0 = x^2 - \frac{1}{3} \\
\|v_3\|^2 &= (v_3, v_3) = \int_{-1}^{1} \left(x^2 - \frac{1}{3} \right)^2 dx = 8/45,
\end{aligned}
$$

Orthogonal basis

- Continued solution:

$$
\begin{aligned}
w_2 &= \frac{v_2}{\|v_2\|} = \frac{x}{\sqrt{2/3}}, \\
v_3 &= u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 \\
&= x^2 - \frac{1}{2} \int_{-1}^{1} x^2 \, dx - \frac{x}{2/3} \int_{-1}^{1} x^3 \, dx \\
&= x^2 - \frac{1}{2} \cdot \frac{2}{3} - \frac{x}{2/3} \cdot 0 = x^2 - \frac{1}{3} \\
\|v_3\|^2 &= (v_3, v_3) = \int_{-1}^{1} \left(x^2 - \frac{1}{3} \right)^2 dx = 8/45, \\
w_3 &= \frac{v_3}{\|v_3\|} = \frac{x^2 - \frac{1}{3}}{\sqrt{8/45}}.
\end{aligned}
$$

Orthogonal basis

- Continued solution: Thus, $\{\frac{1}{\sqrt{2}}, \frac{x}{\sqrt{2/3}}, \frac{x^2-\frac{1}{3}}{\sqrt{8/45}}\}$ is an orthonormal basis of W.

Orthogonal basis

- Continued solution: Thus, $\{\frac{1}{\sqrt{2}}, \frac{x}{\sqrt{2/3}}, \frac{x^2-\frac{1}{3}}{\sqrt{8/45}}\}$ is an orthonormal basis of W.

  And the best approximation in W to $f(x) = e^x \in V$ is

$$Pf = \sum_{j=1}^{3} (f, w_j) w_j$$

Orthogonal basis

- Continued solution: Thus, $\{\frac{1}{\sqrt{2}}, \frac{x}{\sqrt{2/3}}, \frac{x^2-\frac{1}{3}}{\sqrt{8/45}}\}$ is an orthonormal basis of W.

  And the best approximation in W to $f(x) = e^x \in V$ is

$$\begin{aligned} Pf &= \sum_{j=1}^{3} (f, w_j) w_j \\ &= \frac{1}{\sqrt{2}} \int_{-1}^{1} \frac{1}{\sqrt{2}} e^x \, dx + \frac{x}{\sqrt{2/3}} \int_{-1}^{1} \frac{x}{\sqrt{2/3}} e^x \, dx \\ &\quad + \frac{x^2 - \frac{1}{3}}{\sqrt{8/45}} \int_{-1}^{1} \frac{x^2 - \frac{1}{3}}{\sqrt{8/45}} e^x \, dx \end{aligned}$$

Orthogonal basis

- Continued solution: Thus, $\{\frac{1}{\sqrt{2}}, \frac{x}{\sqrt{2/3}}, \frac{x^2-\frac{1}{3}}{\sqrt{8/45}}\}$ is an orthonormal basis of W.

  And the best approximation in W to $f(x) = e^x \in V$ is

$$\begin{aligned} Pf &= \sum_{j=1}^{3} (f, w_j) w_j \\ &= \frac{1}{\sqrt{2}} \int_{-1}^{1} \frac{1}{\sqrt{2}} e^x \, dx + \frac{x}{\sqrt{2/3}} \int_{-1}^{1} \frac{x}{\sqrt{2/3}} e^x \, dx \\ &\quad + \frac{x^2 - \frac{1}{3}}{\sqrt{8/45}} \int_{-1}^{1} \frac{x^2 - \frac{1}{3}}{\sqrt{8/45}} e^x \, dx \\ &\approx 1.1752 + 1.1036x + 0.5367x^2. \end{aligned}$$

Outline

1 Introduction

2 Best approximation

3 Polynomial Approximation

4 Piecewise polynomial approximation

5 Other topics

Basic idea of polynomial approximation

- A simple choice for the subspace W of the best approximation: a set of polynomials!

Basic idea of polynomial approximation

- A simple choice for the subspace W of the best approximation: a set of polynomials!
- Question: can polynomials always provide good approximation?

Basic idea of polynomial approximation

- A simple choice for the subspace W of the best approximation: a set of polynomials!
- Question: can polynomials always provide good approximation?
- Answer: Yes! But we need to be careful about the choices of polynomials.

Basic idea of polynomial approximation

Theorem (I: Weierstrass Approximation Theorem)

Given a function $f \in C[a, b]$ and $\varepsilon > 0$, there exists a polynomial $p(x)$ such that

$$\|p - f\|_\infty = \max_{a \le x \le b} |p(x) - f(x)| \le \varepsilon.$$

Basic idea of polynomial approximation

Theorem (I: Weierstrass Approximation Theorem)

Given a function $f \in C[a, b]$ and $\varepsilon > 0$, there exists a polynomial $p(x)$ such that

$$\|p - f\|_\infty = \max_{a \le x \le b} |p(x) - f(x)| \le \varepsilon.$$

Proof.

See pages 205-209 of the textbooks for the proof and the related materials for the Bernstein polynomials. (Independent study problem) □

Taylor polynomial approximation

Theorem (II: Taylor's expansion)

Suppose that $f \in C^{n+1}[a,b]$ *and* $x_0 \in [a,b]$. *Then for any* $x \in [a,b]$, *we have the following Taylor's expansion of* $f(x)$ *at* x_0:

$$f(x) = P_n(x) + R_n(x),$$

where

$$\begin{aligned} P_n(x) &= \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x-x_0)^k \\ &= f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \cdots \\ &\quad + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n, \end{aligned}$$

Taylor polynomial approximation

Theorem (Continued)

$$R_n = \frac{1}{(n+1)!} f^{(n+1)}(\xi)(x-x_0)^{n+1}$$

for some $\xi \in [x_0, x]$ *(Lagrange form of the remainder)* ,

or

$$R_n = \frac{1}{n!} \int_{x_0}^{x} f^{(n+1)}(s)(x-s)^n \, ds$$

(Integral form of the remainder) .

Lagrange polynomial approximation/interpolation

- What if we don't know $f(x)$ explicitly, but just the values y_j $(j = 0, 1, 2, \cdots, n)$ of $f(x)$ at some points x_j $(j = 0, 1, 2, \cdots, n)$ due to the measurement limitation or the information availability?

Lagrange polynomial approximation/interpolation

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- Basic idea: Given $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and $n+1$ arbitrary numbers $y_0, y_1, \cdots, y_n$, find an interpolating polynomial $p(x)$ of degree at most n such that $y_j = p(x_j)$ $(j = 0, 1, 2, \cdots, n)$.

Lagrange polynomial approximation/interpolation

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- Questions: Does such an interpolating polynomial $p(x)$ exist? If it does, then is it unique?

Lagrange polynomial approximation/interpolation

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- Questions: Does such an interpolating polynomial $p(x)$ exist? If it does, then is it unique?

- Yes!

Lagrange polynomial approximation/interpolation

Theorem (III)

For any $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and for $n+1$ arbitrary numbers $y_0, y_1, y_2, \cdots, y_n$, let

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

Lagrange polynomial approximation/interpolation

Theorem (III)

For any $n+1$ distinct real numbers $x_0, x_1, x_2, \cdots, x_n$ and for $n+1$ arbitrary numbers $y_0, y_1, y_2, \cdots, y_n$, let

$$L_k(x) = \prod_{i=0, i\neq k}^{n} \frac{x - x_i}{x_k - x_i}.$$

Then there exists a unique interpolating polynomial

$$p(x) = \sum_{k=0}^{n} y_k L_k(x)$$

of degree at most n such that $y_j = p(x_j)$ $(j = 0, 1, 2, \cdots, n)$.

Lagrange polynomial approximation/interpolation

Proof:

- First, since each $L_k(x)$ is of degree n, then $p(x)$ is of degree n.

Lagrange polynomial approximation/interpolation

Proof:

- First, since each $L_k(x)$ is of degree n, then $p(x)$ is of degree n.
- Second, it is easy to verify that

$$L_k(x_j) = \delta_{jk} = \begin{cases} 0, & \text{if } j \neq k, \\ 1, & \text{if } j = k. \end{cases}$$

Lagrange polynomial approximation/interpolation

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- Then

$$p(x_j) = \sum_{k=0}^{n} y_k L_k(x_j) = \sum_{k=0}^{n} y_k \delta_{jk} = y_j.$$

Lagrange polynomial approximation/interpolation

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- Then

$$p(x_j) = \sum_{k=0}^{n} y_k L_k(x_j) = \sum_{k=0}^{n} y_k \delta_{jk} = y_j.$$

- Hence we complete the proof of the existence of $p(x)$.

Lagrange polynomial approximation/interpolation

Continued proof:

- Now let's turn to the uniqueness of $p(x)$.

Lagrange polynomial approximation/interpolation

Continued proof:

- Now let's turn to the uniqueness of $p(x)$.
- Assume there are two such polynomials $p(x)$ and $q(x)$ of degree at most n that their values at x_j $(j = 0, 1, 2, \cdots, n)$ are y_j $(j = 0, 1, 2, \cdots, n)$. That is,

$$y_j = p(x_j) = q(x_j),\ j = 0, 1, 2, \cdots, n.$$

Lagrange polynomial approximation/interpolation

Continued proof:

- Now let's turn to the uniqueness of $p(x)$.
- Assume there are two such polynomials $p(x)$ and $q(x)$ of degree at most n that their values at x_j $(j = 0, 1, 2, \cdots, n)$ are y_j $(j = 0, 1, 2, \cdots, n)$. That is,

$$y_j = p(x_j) = q(x_j),\ j = 0, 1, 2, \cdots, n.$$

- Since $p(x)$ and $q(x)$ are polynomials of degree at most n, then we may let

$$\begin{aligned} p(x) &= \sum_{k=0}^{n} \alpha_k x^k, \\ q(x) &= \sum_{k=0}^{n} \beta_k x^k \end{aligned}$$

for some constants α_k and β_k $(k = 0, 1, 2, \cdots, n)$.

Lagrange polynomial approximation/interpolation

Continued proof:

- Let $\gamma_k = \alpha_k - \beta_k$. Then

$$p(x) - q(x) = \sum_{k=0}^{n} (\alpha_k - \beta_k) x^k = \sum_{k=0}^{n} \gamma_k x^k.$$

Lagrange polynomial approximation/interpolation

Continued proof:

- Let $\gamma_k = \alpha_k - \beta_k$. Then

$$p(x) - q(x) = \sum_{k=0}^{n} (\alpha_k - \beta_k) x^k = \sum_{k=0}^{n} \gamma_k x^k.$$

- Since $y_j = p(x_j) = q(x_j)$ $(j = 0, 1, 2, \cdots, n)$, then $p(x_j) - q(x_j) = 0$ $(j = 0, 1, 2, \cdots, n)$.

Lagrange polynomial approximation/interpolation

Continued proof:

- Let $\gamma_k = \alpha_k - \beta_k$. Then

$$p(x) - q(x) = \sum_{k=0}^{n} (\alpha_k - \beta_k) x^k = \sum_{k=0}^{n} \gamma_k x^k.$$

- Since $y_j = p(x_j) = q(x_j)$ $(j = 0, 1, 2, \cdots, n)$, then $p(x_j) - q(x_j) = 0$ $(j = 0, 1, 2, \cdots, n)$.
- Hence

$$\sum_{k=0}^{n} \gamma_k x_j^k = 0, \ j = 0, 1, 2, \cdots, n.$$

Lagrange polynomial approximation/interpolation

Continued proof:

- The above linear system can be rewritten as

$$A \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_n \end{pmatrix} = \begin{pmatrix} 1 & x_0 & x_0^2 & x_0^3 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & x_1^3 & \cdots & x_1^n \\ & \vdots & & \ddots & & \\ 1 & x_n & x_n^2 & x_n^3 & \cdots & x_n^n \end{pmatrix} \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Lagrange polynomial approximation/interpolation

Continued proof:

- The above linear system can be rewritten as

$$A\begin{pmatrix}\gamma_0\\ \gamma_1\\ \vdots\\ \gamma_n\end{pmatrix} = \begin{pmatrix}1 & x_0 & x_0^2 & x_0^3 & \cdots & x_0^n\\ 1 & x_1 & x_1^2 & x_1^3 & \cdots & x_1^n\\ & \vdots & & \ddots & & \\ 1 & x_n & x_n^2 & x_n^3 & \cdots & x_n^n\end{pmatrix}\begin{pmatrix}\gamma_0\\ \gamma_1\\ \vdots\\ \gamma_n\end{pmatrix} = \begin{pmatrix}0\\ 0\\ \vdots\\ 0\end{pmatrix}.$$

- The above matrix A is a Vandermonde matrix and the determinant of the matrxi A is

$$det(A) = \prod_{k=1}^{n}\prod_{j=0}^{k-1}(x_k - x_j).$$

Lagrange polynomial approximation/interpolation

Continued proof:

- The above linear system can be rewritten as

$$A\begin{pmatrix}\gamma_0\\ \gamma_1\\ \vdots\\ \gamma_n\end{pmatrix}=\begin{pmatrix}1 & x_0 & x_0^2 & x_0^3 & \cdots & x_0^n\\ 1 & x_1 & x_1^2 & x_1^3 & \cdots & x_1^n\\ & \vdots & & \ddots & & \\ 1 & x_n & x_n^2 & x_n^3 & \cdots & x_n^n\end{pmatrix}\begin{pmatrix}\gamma_0\\ \gamma_1\\ \vdots\\ \gamma_n\end{pmatrix}=\begin{pmatrix}0\\ 0\\ \vdots\\ 0\end{pmatrix}.$$

- The above matrix A is a Vandermonde matrix and the determinant of the matrxi A is

$$det(A)=\prod_{k=1}^{n}\prod_{j=0}^{k-1}(x_k-x_j).$$

- Since $x_0, x_1, x_2, \cdots, x_n$ are distinct real numbers, then $det(A)\neq 0$.

Lagrange polynomial approximation/interpolation

Continued proof: Recall

Theorem

Let A be an $n \times n$ matrix. Then the following are equivalent:

- *A is nonsingular/invertible.*
- *$det(A) \neq 0$.*
- *$\overrightarrow{x} = 0$ is the unique solution of $A\overrightarrow{x} = 0$.*
- *$A\overrightarrow{x} = \overrightarrow{b}$ has a unique solution.*
- *The columns (and rows) of A are linearly independent.*

Lagrange polynomial approximation/interpolation

Continued proof:

- Hence the above linear system has a unique solution $\gamma_k = 0$ $(k = 1, 2, \cdots, n)$, which implies $\alpha_k = \beta_k$ $(k = 0, 1, 2, \cdots, n)$.

Lagrange polynomial approximation/interpolation

Continued proof:

- Hence the above linear system has a unique solution $\gamma_k = 0$ $(k = 1, 2, \cdots, n)$, which implies $\alpha_k = \beta_k$ $(k = 0, 1, 2, \cdots, n)$.
- Then $p(x) = q(x)$. This completes the proof.

Lagrange polynomial approximation/interpolation

Continued proof:

- Hence the above linear system has a unique solution $\gamma_k = 0$ $(k = 1, 2, \cdots, n)$, which implies $\alpha_k = \beta_k$ $(k = 0, 1, 2, \cdots, n)$.
- Then $p(x) = q(x)$. This completes the proof.

Remark

In fact, the above theory from linear algebra can be used to prove the existence and uniqueness at the same time. (Homework)

Lagrange polynomial approximation/interpolation

- Example 4: Given 3 distinct real numbers $x_0 = 1, x_1 = 2, x_2 = 3$ and 3 numbers $y_0 = -1, y_1 = -2, y_2 = 1$, find the polynomial $p(x)$ such that $p(x_j) = y_j, j = 0, 1, 2$.

Lagrange polynomial approximation/interpolation

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- Solution:

Lagrange polynomial approximation/interpolation

- Example 4: Given 3 distinct real numbers $x_0 = 1, x_1 = 2, x_2 = 3$ and 3 numbers $y_0 = -1, y_1 = -2, y_2 = 1$, find the polynomial $p(x)$ such that $p(x_j) = y_j, j = 0, 1, 2$.
- Solution:

$$L_k(x) = \prod_{i=0, i\neq k}^{2} \frac{x - x_i}{x_k - x_i}, \quad k = 0, 1, 2.$$

Lagrange polynomial approximation/interpolation

- Example 4: Given 3 distinct real numbers $x_0 = 1, x_1 = 2, x_2 = 3$ and 3 numbers $y_0 = -1, y_1 = -2, y_2 = 1$, find the polynomial $p(x)$ such that $p(x_j) = y_j, j = 0, 1, 2$.
- Solution:

$$L_k(x) = \prod_{i=0, i\neq k}^{2} \frac{x - x_i}{x_k - x_i}, \quad k = 0, 1, 2.$$

$$L_0(x) = \prod_{i=0, i\neq 0}^{2} \frac{x - x_i}{x_0 - x_i} = \frac{x - x_1}{x_0 - x_1} \cdot \frac{x - x_2}{x_0 - x_2}$$

$$= \frac{x - 2}{1 - 2} \cdot \frac{x - 3}{1 - 3} = \frac{(x - 2)(x - 3)}{2},$$

Lagrange polynomial approximation/interpolation

$$L_1(x) = \prod_{i=0, i\neq 1}^{2} \frac{x - x_i}{x_1 - x_i} = \frac{x - x_0}{x_1 - x_0} \cdot \frac{x - x_2}{x_1 - x_2}$$
$$= \frac{x-1}{2-1} \cdot \frac{x-3}{2-3} = \frac{(x-1)(x-3)}{-1},$$

Lagrange polynomial approximation/interpolation

$$L_1(x) = \prod_{i=0, i\neq 1}^{2} \frac{x - x_i}{x_1 - x_i} = \frac{x - x_0}{x_1 - x_0} \cdot \frac{x - x_2}{x_1 - x_2}$$

$$= \frac{x-1}{2-1} \cdot \frac{x-3}{2-3} = \frac{(x-1)(x-3)}{-1},$$

$$L_2(x) = \prod_{i=0, i\neq 2}^{2} \frac{x - x_i}{x_2 - x_i} = \frac{x - x_0}{x_2 - x_0} \cdot \frac{x - x_1}{x_2 - x_1}$$

$$= \frac{x-1}{3-1} \cdot \frac{x-2}{3-2} = \frac{(x-1)(x-2)}{2}$$

Lagrange polynomial approximation/interpolation

$$
\begin{aligned}
p(x) &= p_2(x) = \sum_{k=0}^{2} y_k L_k(x) \\
&= y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x) \\
&= -1 \cdot \frac{(x-2)(x-3)}{2} + (-2)\frac{(x-1)(x-3)}{-1} \\
&\quad +1 \cdot \frac{(x-1)(x-2)}{2} \\
&= 2x^2 - 7x + 4.
\end{aligned}
$$

Then we can check that

$$
\begin{aligned}
p(x_0) = p(1) = -1 = y_0, \\
p(x_1) = p(2) = -2 = y_1, \\
p(x_2) = p(3) = 1 = y_2.
\end{aligned}
$$

Lagrange polynomial approximation/interpolation

Definition (Lagrange basis)

The set of functions $\{L_k(x)\}_{k=0}^n$ is called the Lagrange basis for the space of polynomials of degree n associated with the set of points $\{x_k\}_{k=0}^n$. And L_k $(k = 0, 1, 2, \cdots, n)$ are called the Lagrange basis functions.

Lagrange polynomial approximation/interpolation

Definition (Lagrange basis)

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Definition (Lagrange form)

$p(x) = \sum_{k=0}^{n} y_k L_k(x)$ is called the Lagrange form of the interpolating polynomial $p(x)$, which satisfies $y_k = p(x_k)$ $(k = 0, 1, 2, \cdots, n)$.

Lagrange polynomial approximation/interpolation

Remark

What if pick up $y_k = f(x_k)$ $(k = 0, 1, 2, \cdots, n)$ *for a given function* $f(x)$*?*

Lagrange polynomial approximation/interpolation

Remark

What if pick up $y_k = f(x_k)$ $(k = 0, 1, 2, \cdots, n)$ *for a given function* $f(x)$*?*

Definition (Lagrange polynomial approximation)

$p(x) = \sum_{k=0}^{n} f(x_k) L_k(x)$ is called the Lagrange polynomial approximation/interpolation of a given function $f(x)$.

Lagrange polynomial approximation/interpolation

The error of the Lagrange polynomial approximation:

Theorem (IV)

If $x_0, x_1, x_2, \cdots, x_n$ are $n+1$ distinct points in $[a, b]$ and $f \in C^{n+1}[a, b]$, then for each $x \in [a, b]$, there exists a number $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = R(x) = \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$$

where $W(x) = \prod_{i=0}^{n}(x - x_i)$ and $p(x)$ is the Lagrange polynomial approximation/interpolation.

Lagrange polynomial approximation/interpolation

The error of the Lagrange polynomial approximation:

Theorem (IV)

If $x_0, x_1, x_2, \cdots, x_n$ are $n+1$ distinct points in $[a, b]$ and $f \in C^{n+1}[a, b]$, then for each $x \in [a, b]$, there exists a number $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = R(x) = \frac{f^{(n+1)}(\xi(x))W(x)}{(n+1)!}$$

where $W(x) = \prod_{i=0}^{n}(x - x_i)$ and $p(x)$ is the Lagrange polynomial approximation/interpolation.

Proof.

See pages 215-216 of the textbook. (Independent study problem) □

Lagrange polynomial approximation/interpolation

Remark

- *What happen if f is a n-th order polynomial?*
- $f^{(n+1)}(\xi(x)) = 0$ *since* $f^{(n+1)} \equiv 0$.
- *The error* $f(x) - p(x) = 0$.
- *Hence the n-th order Lagrange polynomial approximation is exact for n-th order polynomial.*

Lagrange polynomial approximation/interpolation

Remark

- *Define* $\|f\|_\infty = \max_{a \le x \le b} |f(x)|$. *Then the error bound is*

$$\begin{aligned} \|f - p\|_\infty &= \max_{a \le x \le b} |f(x) - p(x)| \\ &\le \frac{1}{(n+1)!} \left\|f^{(n+1)}\right\|_\infty \|W\|_\infty \\ &\le \frac{1}{(n+1)!} \left\|f^{(n+1)}\right\|_\infty (b-a)^{n+1}. \end{aligned}$$

Lagrange polynomial approximation/interpolation

Remark

- *Define* $\|f\|_\infty = \max_{a\le x\le b} |f(x)|$. *Then the error bound is*

$$\begin{aligned} \|f-p\|_\infty &= \max_{a\le x\le b} |f(x)-p(x)| \\ &\le \frac{1}{(n+1)!}\left\|f^{(n+1)}\right\|_\infty \|W\|_\infty \\ &\le \frac{1}{(n+1)!}\left\|f^{(n+1)}\right\|_\infty (b-a)^{n+1}. \end{aligned}$$

- *In fact, we have*

$$\|f-p\|_\infty \le Ch^{n+1}\left\|f^{(n+1)}\right\|_\infty.$$

where $h = \max_{0\le j\le n-1}(x_{j+1}-x_j)$.

Lagrange polynomial approximation/interpolation

Remark

- *The error bound depends on the nodes* $\{x_i\}_{i=0}^n$ *since* $W(x) = \prod_{i=0}^n (x - x_i)$.

Lagrange polynomial approximation/interpolation

Remark

- *The error bound depends on the nodes* $\{x_i\}_{i=0}^n$ *since* $W(x) = \prod_{i=0}^n (x - x_i)$.
- *Can we choose* $\{x_i\}_{i=0}^n$ *suitably to minimize the error bound?*

Lagrange polynomial approximation/interpolation

Remark

- *The error bound depends on the nodes* $\{x_i\}_{i=0}^n$ *since* $W(x) = \prod_{i=0}^n (x - x_i)$.
- *Can we choose* $\{x_i\}_{i=0}^n$ *suitably to minimize the error bound?*
- *Yes! Chebyshev Polynomials!*

Chebyshev Polynomials approximation/interpolation

Theorem (V)

The uniform norm of $W(x) = \prod_{i=0}^{n}(x - x_i)$ *is minimized on* $[a, b]$ *when*

$$x_i = \frac{1}{2}\left[(b-a)cos\left(\frac{2i+1}{n+1}\cdot\frac{\pi}{2}\right) + a + b\right], \ i = 0, 1, 2, \cdots, n,$$

and the minimum value of the norm is

$$\|W\|_\infty = \frac{1}{2^{2n+1}}(b-a)^{n+1}.$$

Chebyshev Polynomials approximation/interpolation

Theorem (V)

The uniform norm of $W(x) = \prod_{i=0}^{n}(x - x_i)$ *is minimized on* $[a, b]$ *when*

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and the minimum value of the norm is

$$\|W\|_\infty = \frac{1}{2^{2n+1}}(b-a)^{n+1}.$$

Proof.

See pages 217-219 of the textbook for the proof and the related materials on Chebyshev polynomials. (Independent study problem) □

Chebyshev polynomials approximation/interpolation

Definition

If the Lagrange polynomial approximation/interpolation of a function $f(x)$ on $[a, b]$ uses the following roots of Chebyshev polynomials

$$x_i = \frac{1}{2}\left[(b-a)cos\left(\frac{2i+1}{n+1}\cdot\frac{\pi}{2}\right)+a+b\right],\ i = 0, 1, 2, \cdots, n,$$

then it is called the Chebyshev polynomials approximation/interpolation of the function $f(x)$ on $[a, b]$.

Chebyshev polynomials approximation/interpolation

Remark

Compared with the error bound of the regular Lagrange polynomial approximation

$$\|f - p\|_\infty \leq \frac{1}{(n+1)!} \left\|f^{(n+1)}\right\|_\infty (b-a)^{n+1},$$

the error bound of the Chebyshev polynomial approximation is

$$\|f - p\|_\infty \leq \frac{1}{(n+1)!} \left\|f^{(n+1)}\right\|_\infty (b-a)^{n+1} \frac{1}{2^{2n+1}}.$$

Hermit polynomial approximation/interpolation

Definition

For any n distinct real numbers $x_1, x_2, \cdots, x_n$, the Hermit polynomial approximation/interpolation of a given function $f(x)$ is a polynomial $p(x)$ of degree at most $2n-1$ such that

$$p(x_i) = f(x_i), \quad p'(x_i) = f'(x_i), \ i = 1, 2, \cdots, n.$$

Hermit polynomials approximation/interpolation

Theorem (VI)

If $x_1, x_2, \cdots, x_n$ are n distinct real numbers in $[a, b]$ and $f \in C^1[a, b]$. Then there exists a unique Hermit polynomial approximation/interpolation $H_n(x)$ of $f(x)$. And it is given by

$$H_n(x) = \sum_{k=1}^{n} f(x_k) h_k(x) + \sum_{k=1}^{n} f'(x_k) \tilde{h}_k(x),$$

where

$$\begin{aligned} h_k(x) &= \left[1 - 2L_k'(x_k)(x - x_k)\right] (L_k(x))^2, \\ \tilde{h}_k(x) &= (x - x_k) (L_k(x))^2. \end{aligned}$$

Hermit polynomials approximation/interpolation

Theorem (VI: Continued)

Moreover, if $f \in C^{2n}[a, b]$, *then there exists a* $\xi = \xi(x) \in [a, b]$ *such that*

$$f(x) - H_n(x) = \frac{\left(\prod_{k=1}^{n}(x - x_k)\right)^2}{(2n)!} f^{2n}(\xi),$$

Hermit polynomials approximation/interpolation

Theorem (VI: Continued)

Moreover, if $f \in C^{2n}[a, b]$, *then there exists a* $\xi = \xi(x) \in [a, b]$ *such that*

$$f(x) - H_n(x) = \frac{\left(\prod_{k=1}^{n}(x - x_k)\right)^2}{(2n)!} f^{2n}(\xi),$$

Proof.

See page 221 of the textbook. (Independent study problem) □

Some thoughts for the polynomial approximation

Remark

- *In order to obtain a desired accuracy from the polynomial approximation, enough number of interpolating nodes* $\{x_i\}_{i=0}^{n}$ *need to be selected.*

Some thoughts for the polynomial approximation

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- *In order to obtain a desired accuracy from the polynomial approximation, enough number of interpolating nodes* $\{x_i\}_{i=0}^n$ *need to be selected.*

- *When n is large, the corresponding polynomial has very hight order.*

Some thoughts for the polynomial approximation

Remark

- *In order to obtain a desired accuracy from the polynomial approximation, enough number of interpolating nodes* $\{x_i\}_{i=0}^n$ *need to be selected.*
- *When n is large, the corresponding polynomial has very hight order.*
- *Can we use lower order polynomial to achieve high accuracy?*

Some thoughts for the polynomial approximation

Remark

- *In order to obtain a desired accuracy from the polynomial approximation, enough number of interpolating nodes* $\{x_i\}_{i=0}^n$ *need to be selected.*
- *When n is large, the corresponding polynomial has very hight order.*
- *Can we use lower order polynomial to achieve high accuracy?*
- *Yes! Piecewise polynomials!*

Outline

1 Introduction

2 Best approximation

3 Polynomial Approximation

4 Piecewise polynomial approximation

5 Other topics

Basic idea of piecewise polynomial approximation

- Divide the interval $[a, b]$ into many sub-intervals.

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- Continuous piecewise polynomial approximation requires that the piecewise polynomial approximation to be continuous. That is, the polynomial on each sub-interval must match the polynomials on the neighboring sub-intervals.
- Many numerical methods use piecewise polynomial approximation, such as the finite element method.

Continuous piecewise linear approximation

Definition (continuous piecewise linear polynomial)

Given a partition

$$\Delta : a = x_0 < x_1 < x_2 \cdots < x_{n-1} < x_N = b$$

of $[a, b]$, the set L_Δ of all continuous piecewise linear polynomials on $[a, b]$ with respect to Δ is

$$\begin{aligned} L_\Delta \quad = \quad & \{\varphi \in C[a, b] : \ \varphi(x) \text{ is linear on each } [x_i, x_{i+1}] \\ & (i = 0, 1, 2, \cdots, N-1) \text{ of } \Delta\}. \end{aligned}$$

Continuous piecewise linear approximation

Theorem (I)

L_Δ is an $(N+1)-$dimensional subspace of $C[a,b]$.

Proof:

- First, it is easy to verify that L_Δ is a subspace of $C[a,b]$.

Continuous piecewise linear approximation

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L_Δ is an $(N+1)-$dimensional subspace of $C[a,b]$.

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- Consider $\varphi_i(x) \in L_\Delta$ $(i = 0, \cdots, N)$ such that

$$\varphi_i(x_j) = \delta_{ij} = \begin{cases} 0, & \text{if } j \neq i, \\ 1, & \text{if } j = i. \end{cases}$$

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- Let $x = x_j$, then $c_j = 0$ $(j = 0, \cdots, N)$.
- So $\varphi_i(x)$ $(i = 0, \cdots, N)$ are linearly independent.

Continuous piecewise linear approximation

Continued proof:

- Span: Given any $f \in L_\Delta$, let $c_i = f(x_i)$ and consider $g(x) = \sum_{i=0}^{N} c_i \varphi_i(x)$.

Continuous piecewise linear approximation

Continued proof:

- Span: Given any $f \in L_\Delta$, let $c_i = f(x_i)$ and consider $g(x) = \sum_{i=0}^{N} c_i \varphi_i(x)$.
- First, $g(x_j) = c_j = f(x_j)$ $(j = 0, \cdots, N)$.

Continuous piecewise linear approximation

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- First, $g(x_j) = c_j = f(x_j)$ $(j = 0, \cdots, N)$.
- Second, both $f(x)$ and $g(x)$ are linear in each piece $[x_j, x_{j+1}]$ $(j = 0, \cdots, N-1)$.

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Continued proof:

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- First, $g(x_j) = c_j = f(x_j)$ $(j = 0, \cdots, N)$.
- Second, both $f(x)$ and $g(x)$ are linear in each piece $[x_j, x_{j+1}]$ $(j = 0, \cdots, N-1)$.
- Hence $f(x) = g(x)$ in each piece $[x_j, x_{j+1}]$ $(j = 0, \cdots, N-1)$.

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- Hence $f(x) = g(x)$ in each piece $[x_j, x_{j+1}]$ $(j = 0, \cdots, N-1)$.
- Then $f(x) = g(x) = \sum_{i=0}^{N} c_i \varphi_i(x)$.
- This implies $L_\Delta = span\{\varphi_0(x), \cdots, \varphi_N(x)\}$.

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- Then $f(x) = g(x) = \sum_{i=0}^{N} c_i \varphi_i(x)$.
- This implies $L_\Delta = span\{\varphi_0(x), \cdots, \varphi_N(x)\}$.
- Therefore $\varphi_i(x)$ $(i = 0, \cdots, N)$ form a basis of L_Δ if they exist.

Continuous piecewise linear approximation

Continued proof:

- The existence of $\varphi_i(x)$ $(i = 0, \cdots, N)$:

$$\varphi_0(x) = \begin{cases} \frac{x_1 - x}{x_1 - x_0}, & \text{if } x_0 \leq x \leq x_1, \\ 0, & \text{otherwise}, \end{cases}$$

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- A geometric illustration of $\varphi_i(x)$ $(i = 0, \cdots, N)$: see Figure 4.8 on page 240 of the textbook.

Continuous piecewise linear approximation

Remark

- *The continuous piecewise linear basis functions* $\varphi_i(x)$ $(i = 0, 1, 2, \cdots, N)$ *are the well-known hat functions.*

Continuous piecewise linear approximation

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- *The continuous piecewise linear basis functions* $\varphi_i(x)$ $(i = 0, 1, 2, \cdots, N)$ *are the well-known hat functions.*
- *They are actually the linear finite element basis functions for the finite element method in 1D!*

Continuous piecewise linear approximation

Theorem (II)

Given an $f \in C[a,b]$, there is a unique $\Phi \in L_\Delta$ which satisfies $\Phi(x_i) = f(x_i)$ $(i = 0, 1, 2, \cdots, N)$.

Proof:

- Existence: Define $\Phi(x) = \sum_{i=0}^{N} f(x_i)\varphi_i(x)$.

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- Then $\Phi_1 - \Phi_2 = 0$ in $[x_j, x_{j+1}]$ $(j = 0, \cdots, N-1)$.
- Hence $\Phi_1(x) = \Phi_2(x) \Rightarrow$ uniqueness of Φ.

Continuous piecewise linear approximation

Definition (Interpolation)

$\Phi(x) = \sum_{i=0}^{N} f(x_i)\varphi_i(x)$ is called the L_Δ interpolation of f, which is denoted by $I_N f(x)$.

Continuous piecewise linear approximation

Definition (Interpolation)

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Remark

- $I_N : C[a,b] \to L_\Delta$ *is a linear operator, i.e.,*

$$I_N(a_1 f_1(x) + a_2 f_2(x)) = a_1 I_N f_1(x) + a_2 I_N f_2(x).$$

Continuous piecewise linear approximation

Theorem (III)

If $f \in C^2[a,b]$, *then*

$$\begin{aligned} \|f - I_N f\|_\infty &\leq \frac{1}{8} h^2 \|f''\|_\infty , \\ \|(f - I_N f)'\|_\infty &\leq \frac{1}{2} h \|f''\|_\infty . \end{aligned}$$

If $f \in C^1[a,b]$, *then*

$$\|f - I_N f\|_\infty \leq \frac{1}{2} h \|f'\|_\infty ,$$

where $h = \frac{b-a}{N}$.
If $f \in C[a,b]$, *then*

$$\|f - I_N f\|_\infty \to 0 \quad as \quad N \to \infty.$$

Continuous piecewise linear approximation

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$$\|f - I_N f\|_\infty \to 0 \quad as \quad N \to \infty.$$

Proof: See pages 241-242 of textbook (Independent study problem).

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1 Introduction

2 Best approximation

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5 Other topics

Topics: Independent study problems

- Newton form of the Lagrange polynomials (section 4.3.4)
- Least square approximation (section 4.3.6 and 4.3.7)
- Minmax approximation (section 4.3.8)
- Interval bounds on the errors (section 4.3.9)
- Cubic spline interpolation and B-splines (section 4.4.2 and 4.4.3).
- Trigonometric Approximation (section 4.5)
- Rational approximation (section 4.6)
- Wavelet (section 4.7)
- .....